Learning normalized image densities via dual score matching

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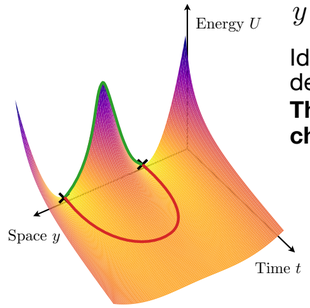
Q: Diffusion models generate great samples, but $p_\theta(x)$ is implicit. Can we make it explicit? What are its properties?

We propose a novel objective + architecture to learn log probabilities.
Surprising observations: **no concentration on an "image manifold"!**

Methods

Objective: dual score matching

Learn a time-dependent energy along the diffusion: $p_\theta(y|t) = \frac{1}{Z_\theta} \mathrm{e}^{-U_\theta(y,t)}$



$y =$ clean data x + noise of variance t

Idea: parameterize the energy but constrain **both** derivatives (dual score matching).
The network learns to "integrate" energy changes along the diffusion.

Data score matching: (Tweedie/Miyasawa)

$$\ell_{\mathrm{DSM}}(\theta,t) = \mathbb{E}_{x,y}\left[\|\nabla_y U_\theta(y,t) - \frac{y-x}{t}\|^2\right]$$

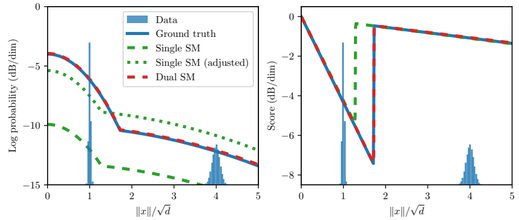
Time score matching:

$$\boxed{\ell_{\mathrm{TSM}}(\theta,t) = \mathbb{E}_{x,y}\left[\left(\partial_t U_\theta(y,t) - \frac{d}{2t} + \frac{\|y-x\|^2}{2t^2}\right)^2\right]}$$

Post-training normalization: almost Gaussian at large t!

Dual SM recovers ground truth even on high-dim multimodal data: (while single SM fails)

Data: scale mixture of two Gaussians in $d = 1000$



Architecture: from score to energy

Goal: preserve inductive biases of a baseline score network $s_\theta(y,t)$
Possible if it is homogeneous: $s_\theta(y,t) = \nabla_y s_\theta(y,t) y$

$$\boxed{U_\theta(y,t) = \frac{1}{2}\langle y, s_\theta(y,t)\rangle} \longrightarrow \nabla_y U_\theta(y,t) = \frac{1}{2}\left(\nabla_y s_\theta(y,t) + \nabla_y s_\theta(y,t)^\top\right) y$$

Results

We train an energy network derived from a UNet on ImageNet64 with dual score matching. Denoising/sampling performance is same or better than score network.

Generalization

We trained two models on disjoint sets of N samples. For large N, **they learn the same probability model!**

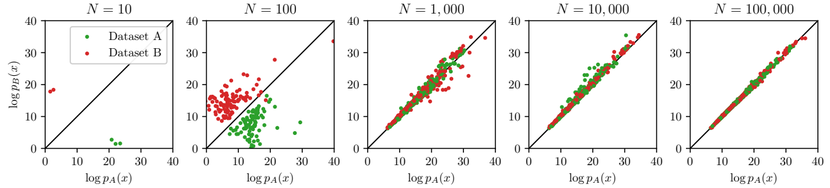
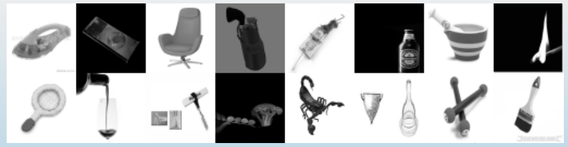


Image content and probability

Highest probability test images:

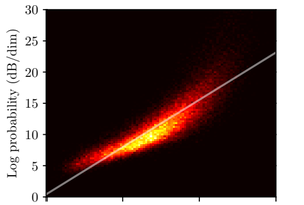


Lowest probability test images:

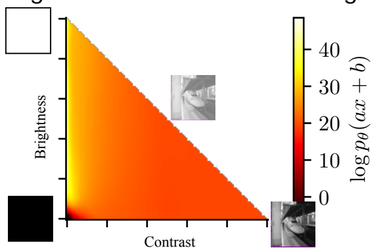


They have a probability ratio of $10^{14,000}$!

Probability is correlated with sparsity of image content:



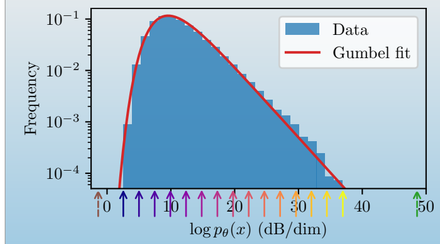
Decreasing contrast increases probability: every image connected to constant images



Distribution of probabilities

Differential entropy: -11.4 dB/dim ($\approx$ volume of $[0,0.1]^d$ compared to $[0,1]^d$).
With 256 grayscale levels, $10^{5,180}$ natural images out of $10^{9,860}$ total images.

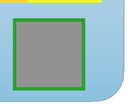
Distribution of log probabilities, test set:





- Broad asymmetrical distribution (no concentration!)
- Higher probability ≠ more frequent



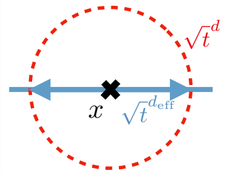


Local dimensionality (testing manifold hypothesis)

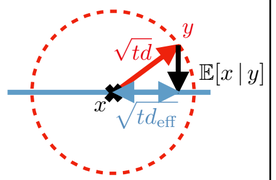
Two **equivalent** dimensionality definitions:

$$\boxed{\begin{aligned} d_{\mathrm{eff}}(x,t) &= d + 2t\partial_t \mathbb{E}_y[\log p_\theta(y,t)\,|\,x] \\ &= \frac{1}{t}\mathbb{E}_y[\|x - \mathbb{E}_x[x\,|\,y]\|^2\,|\,x] \end{aligned}}$$

mass/volume growth

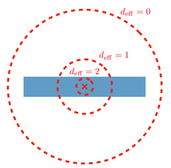


denoising as projection

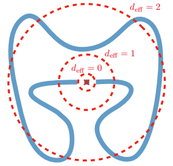


Dimensionality depends on scale:

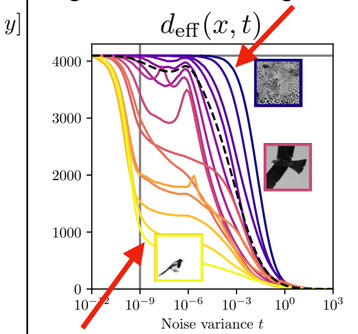
decreasing



increasing



Big full-dimensional regions



Tiny low-probability ridges

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